

String Properties:

1] Immutability - Once String is created, it cannot be modified. String is immutable.
eg. Str1 = "hello"

X modified

add

delete

replace

hello world

hello

Hello

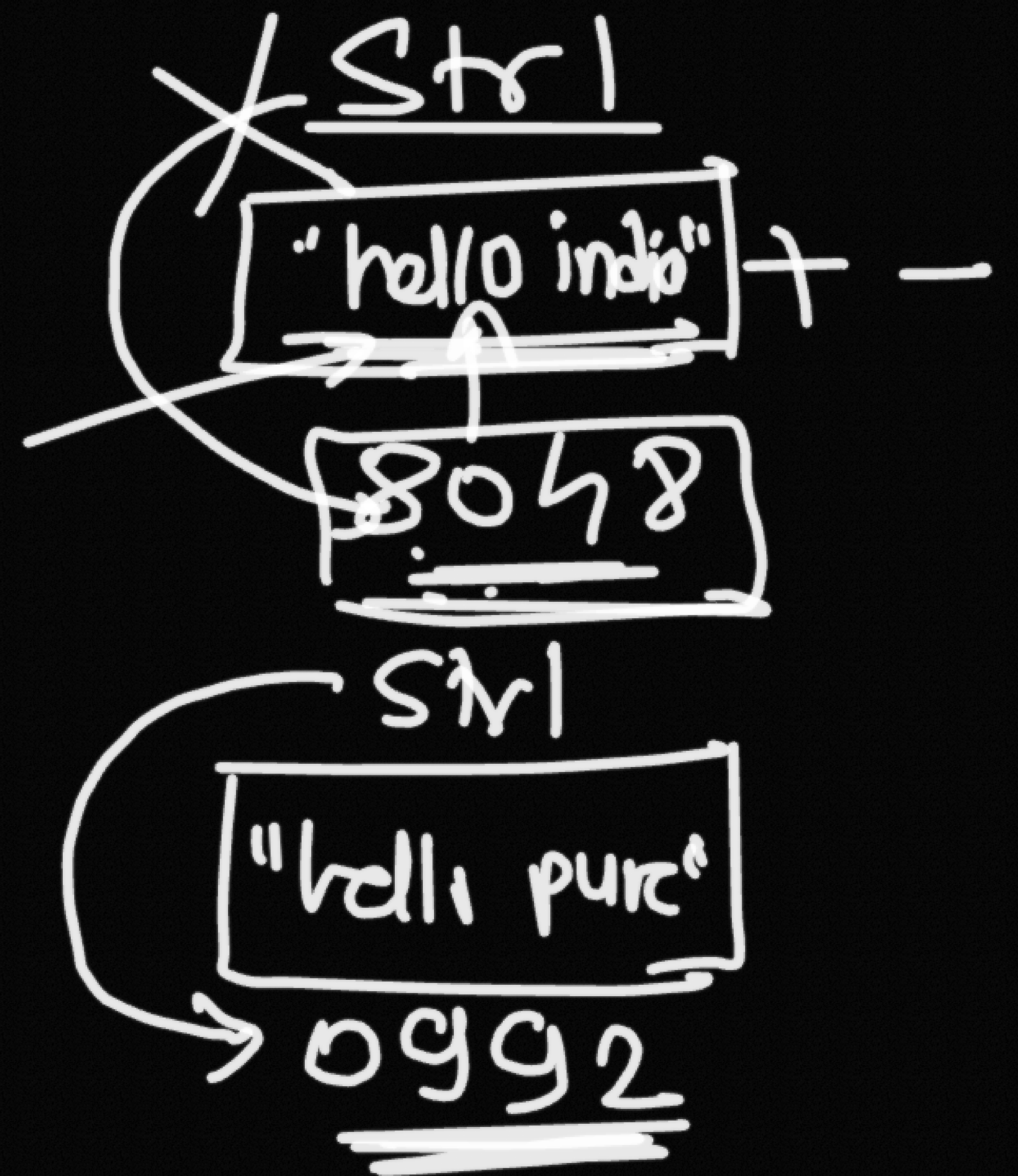
Str1 = "hello india"

Print (Str1) → "hello india"

String is immutable

Str1 = "hello pune"

Print (Str1)



2] Concatenation (+) - we use this '+' operator to combine two or more strings.

eg. greeting = "Hello" ←
name = "Garish" ←

message = greeting + name
"Hello" + "Garish"

Print(message) → Hello Garish

3] Repetition — (*)

* operator repeats string multiple time.

```
print ("Hi")
```

```
print ("Hi")
```

```
print ("Hi")
```

o/p- Hi Hi Hi

word = "Hi"

repeated_word = word * 3

print(repeated_word)

o/p - Hi Hi Hi

4] Membership Testing (in or not in)

- checks if substring (part of string) exists in string
- if substring exists it returns
True
- if does not exist it returns
False

eg. text = "Python programming"

print("python" in text) → True

print("Java" in text) → False

Built in functions in strings

1) `len()` — It returns the length of string.

length → No. of chars in the string

name = "ganes₁h₂s₃s₄s₅" → length → 6

`Print(len(name))` → 6

2] min() - It returns the smallest characters from the string based on ASCII code

name = ⁽⁷¹⁾g ⁽⁹⁷⁾a ¹¹⁰n ¹⁰¹e ¹¹⁵s ¹⁰⁴h "

~~step~~ ^G Print (min(name))

o/n - a

3] max() - It returns
largest character from the
string based on ASCII code

eg. name = "AKshay"
 65 67 115 104 47 121
print(max(name))

o/p- y

- Built-in - functions

Syntax - fun_name (String_var_name)

eg. len(), min(), max()

eg. len(name)

- String methods :-

Syntax : String_var_name . method_name()

eg. upper() - name.upper()

String methods :-

1] Content testing methods

1. isalpha () - It checks if all characters are alphabets.

alphabets - $\frac{A-Z}{0-9}$, $\frac{a-z}{@ \$ \%}$

greetings = "Hello world"

Print(greetings.isalpha())

- if all characters are alphabets it returns True
- if all characters are not alphabets it returns False

2. `isdigit()` — it checks if all characters are digits.

eg. `Str1 = "12345"`
 ↓ ↓ ↓ ↓ ↓
`Print (Str1. isdigit())` → True

`Str2 = "12345a"`
 ↓ ↓ ↓ ↓ ↓ x
`Print (Str2. isdigit())` → False

3. `isalnum()` - checks if all characters are alphabets or digits

eg. `str4 = "Hello123"`
 ✓✓✓✓✓✓✓

`print(str4.isalnum())`
 → True

